

Batch Code: ANP_D1838

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LAB-05

Question:

1. Write a java program using string to find number of uppercase, lowercase, digits using string method

input - HeLLO ChENNAI weLCOME 600024

2. Write a java program using string to remove duplicate words in the given string

input - Welcome to chennai city and welcome to beautiful city

output - Welcome to chennai city and beautiful

3. Write a difference between String, StringBuilder and StringBuffer
-

1. Java Program to Count Uppercase, Lowercase, and Digits.

Program:

```
public class CountCharUsingString {  
  
    public static void main(String[] args) {  
  
        String str = "HeLLO ChENNAI weLCOME 600024";  
  
        int UpperCase=0;  
        int LowerCase=0;  
        int Digits=0;  
        int Space=0;  
        int i=0;
```

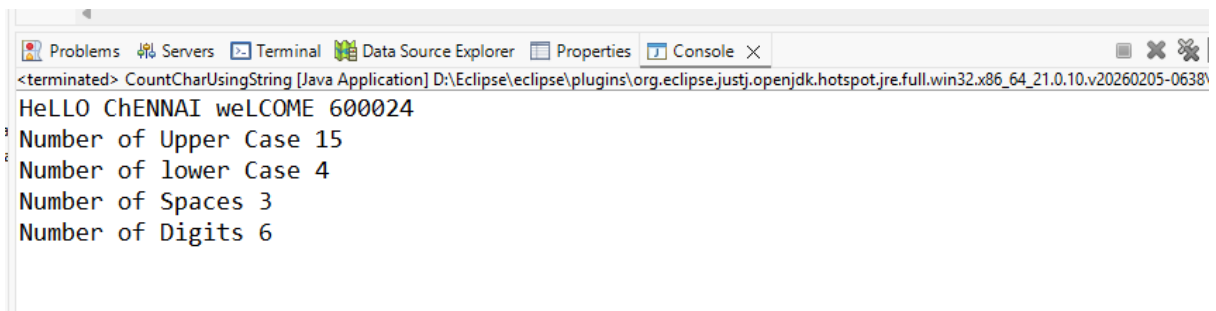
```

for(i=0; i<str.length(); i++)
{
    Character ch = str.charAt(i);
    System.out.print(ch);

    if(ch>='A' && ch<='Z')
    {
        UpperCase++;
    }
    else if(ch>='a' && ch<='z')
    {
        LowerCase++;
    }
    else if(ch==' ')
    {
        Space++;
    }
    else if(ch>='0' && ch<='9')
    {
        Digits++;
    }
}
System.out.println("\nNumber of Upper Case " + UpperCase);
System.out.println("Number of lower Case " + LowerCase);
System.out.println("Number of Spaces " + Space);
System.out.println("Number of Digits " + Digits);
}
}

```

Output:



```

<terminated> CountCharUsingString [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\
HeLLO ChENNAI weLCOME 600024
Number of Upper Case 15
Number of lower Case 4
Number of Spaces 3
Number of Digits 6

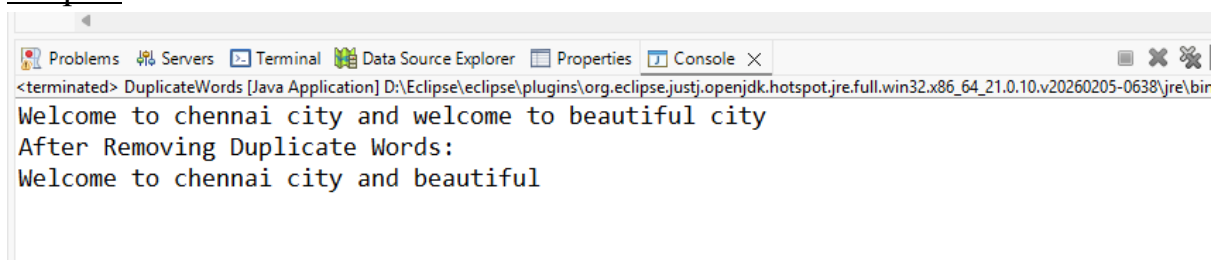
```

2. Java Program to Remove Duplicate Words.

Program:

```
public class DuplicateWords {  
    public static void main(String[] args) {  
  
        String str = "Welcome to chennai city and welcome to beautiful city";  
        System.out.println(str);  
  
        String[] words = str.split(" ");  
        String result = "";  
  
        for (int i = 0; i < words.length; i++) {  
            boolean duplicate = false;  
  
            for (int j = 0; j < i; j++) {  
                if (words[i].equalsIgnoreCase(words[j])) {  
                    duplicate = true;  
                    break;  
                }  
            }  
  
            if (!duplicate) {  
                result += words[i] + " ";  
            }  
        }  
        System.out.println("After Removing Duplicate Words:");  
        System.out.println(result.trim());  
    }  
}
```

Output:

A screenshot of the Eclipse IDE's console window. The window title bar shows 'Problems', 'Servers', 'Terminal', 'Data Source Explorer', 'Properties', and 'Console'. The console output shows the execution of the Java program. It starts with the string 'Welcome to chennai city and welcome to beautiful city' being printed. Then, it prints 'After Removing Duplicate Words:' followed by the result 'Welcome to chennai city and beautiful', which is the original string with the duplicate word 'city' removed.

```
<terminated> DuplicateWords [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre\bin  
Welcome to chennai city and welcome to beautiful city  
After Removing Duplicate Words:  
Welcome to chennai city and beautiful
```

3. Difference Between String, StringBuilder and StringBuffer.

Feature	String	StringBuilder	StringBuffer
Definition	A sequence of characters stored as an object.	A mutable sequence of characters.	A mutable sequence of characters with thread safety.
Mutability	Immutable (cannot be changed).	Mutable (can be changed).	Mutable (can be changed).
Memory Allocation	Creates a new object whenever modified.	Modifies the existing object.	Modifies the existing object.
Performance	Slow for frequent modifications.	Fastest among the three.	Slower than StringBuilder because of synchronization.
Execution Speed	Slow	Fast	Moderate
Heap Memory Usage	More memory consumption.	Less memory consumption.	Less memory consumption.
Suitable For	Fixed text data.	Single-threaded applications.	Multi-threaded applications.
Package	java.lang.String	java.lang.StringBuilder	java.lang.StringBuffer
Append Operation	Creates new object.	Changes existing object.	Changes existing object.

// Important Methods in Program:

Length()	Find string length
CharAt(i)	Get character by index
split(" ")	Split string into words
equalsIgnoreCase()	Compare strings ignoring case
trim()	Remove extra spaces